CHEONGJAE LEE
Local Address: **Boston, MA** | Permanent Address: Uiwang-si, Gyeonggi-do, Republic of Korea
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EDUCATION

BOSTON UNIVERSITY — Boston, MA
Master's in Electrical Engineering — Expected Graduation: 12/27
Main Focus of study: ***Cyber-Physical Systems, Robotics, Computer Vision***

THE PENNSYLVANIA STATE UNIVERSITY — State College, PA
Bachelor of Science in ***Electrical Engineering - GPA: 3.19/4.0*** — 12/23
Minor in ***Engineering Mechanics*** — Degree Certification Credential ID: 24EA-OU1S-CHEI
CeDiploma Validation: https://www.registrar.psu.edu/graduation/diplomas/cediploma/validate

PROFESSIONAL EXPERIENCE

SOELECT INCORPORATED — Greensboro, NC
Manufacturing Process Engineer — 03/26 – 05/26

- Operated and monitored roll-to-roll equipment for lithium metal anode manufacturing.
- Evaluated anode quality using optical microscopy, electrolyte contact-angle measurements, peel-strength testing, and SEM/EDX characterization.
- Assembled ~5mAh pouch cells manually, integrating cathodes, separators, lithium metal anodes, and liquid electrolyte.
- Assessed electrochemical performance through electrochemical impedance spectroscopy (EIS), cycling tests, and cell teardown analysis.

FASTLION ENERGY INCORPORATED — State College, PA
Electrical Engineer — **0**7/24 – 03/26

- Designed and prototyped lithium-ion Battery Management Systems for 6T military and industrial battery platforms, supporting cell monitoring, fault detection, and system-level protection.
- Developed and validated CAN communication protocols for battery telemetry, diagnostics, and control integration with vehicle and system controllers.
- Designed multi-layer PCBs supporting PWM control, ultra-precision analog sensing, digital communication interfaces (SPI, I2C), and high-current power paths.
- Collaborated cross-functionally with mechanical and systems engineers to ensure electrical designs met performance, safety, and environmental requirements.
- Supported bring-up, debugging, and validation of prototype hardware using oscilloscopes, logic analyzers, and bench instrumentation.

EC POWER GROUP — State College, PA
Electrical Engineer — **0**2/24 – 07/24

- Designed and programmed microcontroller-based systems for Battery Management System prototyping, including embedded firmware development in C/C++.
- Led end-to-end development of a custom cell-level battery test platform, encompassing PCB design, embedded software, and data acquisition workflows.
- Implemented sensor interfaces and control logic to support accurate electrical characterization and performance testing of battery cells.
- Collaborated with senior engineers to iterate rapidly on hardware designs and testing methodologies.

Electrical Engineering Intern — **0**5/23 – 02/24

- Designed, assembled, and debugged electronic circuits and early-stage prototypes supporting battery development efforts.
- Developed controller firmware and PCB layouts for a battery test rig, enabling automated testing and data capture.
- Assisted in battery manufacturing and test operations, gaining exposure to production-adjacent engineering workflows.

SKILLS: Python | C/C++ | LabVIEW | MATLAB | SolidWorks | Altium | KiCAD | Abaqus | Linux | Isaac Sim | ROS2